

Designability of RNA Targets with Up to Two Length-2 Helices

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Abstract

RNA inverse folding asks whether one can choose an RNA sequence whose unique optimal secondary structure is a prescribed target. We study this problem in a simplified four-letter model: only Watson–Crick A–U and C–G pairs are allowed, structures contain no pseudoknots, adjacent nucleotides may pair, and lower energy means a larger number of base pairs.

Earlier work showed that a large class of targets is designable when every helix contains at least three base pairs. Shorter helices are more difficult because they provide less freedom for coordinating the base-pair choices made at neighboring loops. We prove that designability is nevertheless guaranteed when the target contains at most two helices of length 2, contains no helix of length 1, avoids the two standard local obstruction motifs m_5 and $m_{3\bullet}$, and has all remaining helices of length at least 3.

The proof labels the target base pairs with three colors that encode how nucleotides will later be assigned. A length-2 helix can require a flexible gray connection at an adjacent loop. Because the target contains at most two such helices, no loop must accommodate more than two of these demands. When both demands meet at the same multiloop, the incoming longer helix can be colored to end with a non-gray pair, leaving the two available gray choices for the short branches. These local choices can then be assembled consistently across the entire target.

From the resulting coloring we construct an explicit RNA sequence and prove that every different compatible fold has fewer base pairs than the target. The complete theorem and its supporting lemmas were formalized in Lean 4 and independently rebuilt from frozen source in an isolated environment. The work was developed with substantial AI assistance under human supervision. At the time of this release, the manuscript and formal proof have not yet been reviewed by an independent human subject-matter expert.

Keywords: RNA inverse folding; RNA design; secondary structure; separated coloring; isolated stack; formal verification; Lean.

1 Introduction

1.1 RNA inverse folding

RNA inverse folding starts from a target secondary structure and asks for a nucleotide sequence that folds uniquely into that target under a specified energy model. This reverses the usual structure-prediction problem: the structure is fixed, while the sequence must be designed. Representative approaches include ViennaRNA’s RNAinverse, a canonical early thermodynamic inverse-folding heuristic; the stochastic local search of RNA-SSD; INFO-RNA’s dynamic-programming initialization followed by local search; the constraint-programming method RNAiFold; and NUPACK’s ensemble-defect optimization [1, 6, 7, 9, 13]. Bonnet et al. proved NP-completeness for a Watson–Crick design-extension

formulation with prescribed nucleotide constraints [2], so exact positive results for well-defined structural classes remain useful.

We study the combinatorial Watson–Crick base-pair model studied by Haleš et al. [8]. Every permitted base pair contributes the same energy, and loops carry no additional energy term. A sequence is a design for a target when the target is its unique compatible noncrossing structure with the maximum number of base pairs. This is an idealized model, not a replacement for Turner nearest-neighbor thermodynamics [12]. Its advantage is that structural conditions can be proved exactly and formalized completely.

1.2 Separated colorings and prior work

Haleš et al. represented a secondary structure by a rooted ordered tree whose paired nodes are colored black, white, or gray. They defined proper and separated colorings and proved that a separated coloring yields a design in the four-letter Watson–Crick model [8]. They also identified the local obstructions m_5 and $m_{3\bullet}$ and characterized several designable classes.

Boury, Bulteau, and Ponty introduced modulo- m separation and proved motif-free separability hardness [3]. Their 2025 journal expansion strengthens the hardness result to targets without isolated base pairs [4]. They also proved a linear-time positive result for targets in which every helix has length at least three. In their terminology, a helix of length 1 is an isolated base pair and a helix of length 2 is an isolated stack. Their known $h_{\min} = 2$ construction uses many isolated stacks. This motivates asking whether a bounded number of isolated stacks admits a general positive theorem.

1.3 Why isolated stacks require an additional argument

A helix of length at least three has enough positions to adjust the parity of its terminal level while preserving the color of its first pair. A length-2 helix has fewer valid color patterns. With two isolated stacks, two child subtrees may both require a gray first pair at the same multiloop. If the incoming helix also ends gray, the loop would contain three gray incidences, while properness allows only two.

The construction below avoids that conflict. If two child subtrees already contain the two isolated stacks allowed in the target, then the incoming helix cannot itself be short. It is therefore long enough to reach the required terminal parity while ending black or white rather than gray. The two gray colors can then be assigned to the two child helices.

1.4 Main result and proof outline

The paper makes four main contributions.

1. **At-most-two theorem.** Every $(m_5, m_{3\bullet})$ -free target with no isolated base pair, at most two isolated stacks, and all remaining helices of length at least three admits a modulo-2 separated coloring and is designable in the Watson–Crick base-pair model.
2. **Induction on helix subtrees.** Each helix subtree is summarized by the entry states it is guaranteed to support. The sets F and Q distinguish subtrees without an isolated stack from those containing at least one.
3. **Self-contained design proof.** We include the color-to-sequence construction, the nucleotide-counting bound, uniqueness on the paired restriction, and the exclusion of alternative optimal folds.
4. **Formal verification.** The complete theorem was formalized in Lean 4. The frozen source was rebuilt in isolation, and a blinded automated AI audit separately compared the formal statement with an anonymous mathematical specification.

The theorem is a sufficient condition, not a characterization of all designable targets with isolated stacks. The target $((.))(.(.))(.(.))$ admits no proper modulo-2 separated coloring at all. It nevertheless has a proper ordinarily separated coloring and is uniquely designable, so the example marks a sharp boundary for this modulo-2 criterion rather than a general designability boundary.

1.5 AI-assisted proof development

The proof was developed with extensive assistance from ChatGPT, Codex, and Claude Code under human supervision. The human author fixed the scientific model, clarified the quantifiers, selected the acceptance criteria, and controlled the canonical versions. AI systems proposed arguments, generated test programs and Lean code, and were also used in separate sessions to search for counterexamples and hidden assumptions. Several early formulations were rejected or repaired before the final proof was frozen.

The evidence is ranked rather than treated as a vote among AI systems. In descending order of logical strength, it consists of Lean kernel acceptance of the exact final theorem; isolated clean reconstruction of the frozen source and pinned dependencies; source-integrity, dependency, and axiom audits; a blinded automated source-to-specification review by a fresh AI session; and independent finite computational checks and negative controls. The correctness claim does not rest on agreement among AI systems. Its strongest evidence is kernel checking of the formal proof and isolated reconstruction of the frozen source. The AI reviews are adversarial checks of interpretation, model fidelity, and exposition. Section 14 records these checks and their limitations.

Figure 1 separates the new coloring construction from the established coloring-to-design implication and from the three layers of formal-artifact checking.

1.6 Organization

Sections 2–3 define the model and summarize prior results. Sections 4–8 construct the coloring. Sections 9–12 convert it into a unique design. Section 13 gives examples, Section 14 describes the Lean development and artifact checks, Section 15 records the AI-assisted development and audit trail, Section 16 discusses scope and open questions, and Section 17 concludes.

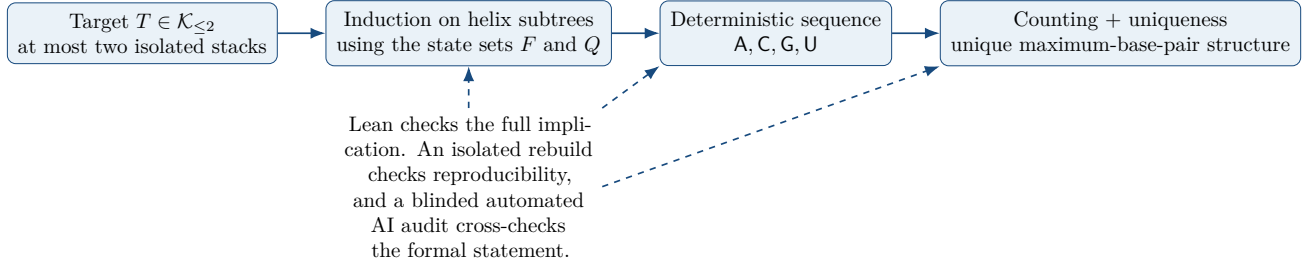


Figure 1: Structure of the proof. The new step is the construction of a proper modulo-2 separated coloring for targets with at most two isolated stacks. The remaining implications are proved self-containedly and formalized in Lean.

2 Model and definitions

2.1 Sequences, structures, and energy

Fix $n \geq 0$ and let $[n] = \{1, \dots, n\}$. An RNA sequence is a word $w \in \{A, C, G, U\}^n$. The Watson–Crick complement involution is

$$\text{comp}(A) = U, \quad \text{comp}(U) = A, \quad \text{comp}(C) = G, \quad \text{comp}(G) = C.$$

An ordered pair of nucleotides is compatible precisely when the second is the complement of the first. G–U wobble is not permitted.

A secondary structure P on $[n]$ is a set of arcs (i, j) satisfying:

1. $1 \leq i < j \leq n$;
2. each position belongs to at most one arc; and
3. the arcs are noncrossing: no two arcs $(i, j), (k, l)$ satisfy $i < k < j < l$.

We impose minimum arc length $\theta = 0$, so adjacent positions may pair. A structure is compatible with w if every arc joins complementary letters. Its energy is

$$E(w, P) = -|P|.$$

Thus lower energy is exactly larger pair count.

Definition 2.1 (Design and designability). *A sequence w is a design for a target T if T is compatible with w and every distinct compatible noncrossing structure S on the same sequence satisfies*

$$|S| < |T|.$$

A target is designable if it admits a design.

The quantifier order is important: one sequence is chosen first, and then every compatible noncrossing structure of that same sequence is considered. Competitors are not required to share the target’s stems, helix decomposition, tree, or number of pairs.

2.2 Tree representation

Associate to a structure P the rooted ordered tree representation used by Haleš et al.; we also refer to it as the interval tree. The nodes are:

- a virtual root $r = [0, n + 1]$;
- a paired node $[i, j]$ for each arc $(i, j) \in P$; and
- an unpaired singleton $[k, k]$ for every position not incident to an arc.

The children of an interval are its maximal proper subintervals in backbone order. Noncrossingness makes the family laminar, so every nonroot node has a unique smallest enclosing paired interval or the virtual root.

For a paired node v , let $d(v)$ be its number of paired children. Its paired degree is

$$\deg(v) = 1 + d(v) \quad \text{for nonroot paired } v, \quad \deg(r) = d(r).$$

The root has no parent contribution, and unpaired children do not contribute to paired degree.

2.3 Forbidden motifs

The two standard local obstructions are:

- m_5 : some node has paired degree greater than 4;
- $m_{3\bullet}$: some node has an unpaired child and paired degree greater than 2.

Lemma 2.2 (Motif bounds). *If a target avoids m_5 and $m_{3\bullet}$, then:*

<i>node</i>	<i>has an unpaired child?</i>	<i>maximum paired children</i>
<i>nonroot paired</i>	<i>yes</i>	1
<i>nonroot paired</i>	<i>no</i>	3
<i>root</i>	<i>yes</i>	2
<i>root</i>	<i>no</i>	4

Proof. For a nonroot paired node, $\deg = 1 + d$. If it has an unpaired child, avoiding $m_{3\bullet}$ gives $1 + d \leq 2$; otherwise avoiding m_5 gives $1 + d \leq 4$. The root has degree d , giving the remaining rows. $\square$

2.4 Helices and the target class

A helix of length h is a maximal consecutive nested run

$$(i, j), (i + 1, j - 1), \dots, (i + h - 1, j - h + 1).$$

A length-1 helix is an isolated base pair; a length-2 helix is an isolated stack. Maximality is required both outward and inward.

Lemma 2.3 (Helix chains). *Two pairs $[i, j]$ and $[i + 1, j - 1]$ are consecutive in a helix if and only if $[i, j]$ has exactly one paired child, namely $[i + 1, j - 1]$, and has no unpaired child.*

Proof. If the pairs are stacked, there are no backbone positions available for another child between their corresponding endpoints. Conversely, if $[i, j]$ has one paired child $[k, l]$ and no unpaired child, its children cover all positions strictly inside (i, j) ; hence $k = i + 1$ and $l = j - 1$. $\square$

The consecutive-pair relation partitions the target pairs into disjoint maximal chains, so every target pair belongs to exactly one maximal helix.

Definition 2.4 (At-most-two target class). *Let $\mathcal{K}_{\leq 2}$ be the class of targets satisfying:*

1. at most two maximal helices have length 2;
2. no maximal helix has length 1;
3. every maximal helix whose length is not 2 has length at least 3;
4. m_5 is absent; and
5. $m_{3\bullet}$ is absent.

The all-unpaired target belongs to $\mathcal{K}_{\leq 2}$ vacuously.

Condition 2 is stated explicitly to keep the scientific boundary visible and to mirror the frozen Lean predicate. Logically it follows from condition 3, because a helix of length 1 has length different from 2 but cannot have length at least 3.

Remark 2.5 (Terminology). *We follow the terminology of Haleš et al. and Boury et al.: tree representation, proper coloring, level, separated coloring, isolated base pair, and isolated stack. The symbols E , L , and M are only local abbreviations for the three terminal-loop types defined below. The symbols F and Q name two explicitly listed sets of feasible entry states; they are not additional biological objects.*

2.5 Colors, exposures, and levels

Color every nonroot paired node black B, white W, or gray G. Define

$$\text{inv}(\text{B}) = \text{W}, \quad \text{inv}(\text{W}) = \text{B}, \quad \text{inv}(\text{G}) = \text{G},$$

and

$$\delta(\text{B}) = +1, \quad \delta(\text{W}) = -1, \quad \delta(\text{G}) = 0.$$

For a nonroot paired node v , its exposed multiset is

$$X(v) = \{\text{inv}(c(v))\} \uplus \{c(u) : u \text{ is a paired child of } v\}.$$

At the root,

$$X(r) = \{c(u) : u \text{ is a paired child of } r\}.$$

Definition 2.6 (Proper coloring). *A coloring is proper if every exposed multiset contains at most one black, at most one white, and at most two gray incidences.*

At the root, properness is exactly the capacity bound on its paired children: at most one black, one white, and two gray. At a nonroot black node the child capacities are one black, no white, and two gray; at a white node they are no black, one white, and two gray; and at a gray node they are one of each color. Equivalently, away from the root, a node has at most one child of its own color, black and white are never parent–child neighbors, and the remaining color capacities above also hold. This is the properness formulation used in the separated-coloring literature [3, 8].

The inclusive level of a paired node is

$$\ell(v) = \sum_{u \in r \rightsquigarrow v} \delta(c(u)),$$

where the sum runs over the colored nonroot paired nodes on the path from the root to v , including v . The root level is zero. An unpaired child of a paired node inherits the parent’s level; an unpaired root child has level zero. The entry level of a nonroot paired node is

$$e(v) = \ell(\text{parent}(v)) = \ell(v) - \delta(c(v)).$$

Definition 2.7 (Separated and modulo-2 separated). *A proper coloring is separated if no integer level is occupied both by a gray paired node and by an unpaired node. Following Boury et al., a coloring is modulo-2 separated if the residues of gray paired nodes and unpaired nodes are disjoint modulo two. Our construction yields the more specific situation in which there are residues $\xi, \eta \in \mathbb{Z}/2\mathbb{Z}$, with $\eta = 1 - \xi$, such that every unpaired node has residue ξ and every gray paired node has residue η . We call this the uniform residue form of modulo-2 separation. It is the property named **StrongTwoSeparated** in the Lean development.*

Lemma 2.8. *Every coloring in the uniform residue form is separated.*

Proof. The gray and unpaired level sets lie in different parity classes. $\square$

Lemma 2.9 (Nonvacuous modulo-2 separation has uniform residue form). *If a proper modulo-2 separated coloring has at least one unpaired node and at least one gray paired node, then there are distinct residues $\xi, \eta \in \mathbb{Z}/2\mathbb{Z}$ such that all unpaired levels have residue ξ and all gray-pair levels have residue η .*

Proof. The two residue sets are nonempty and disjoint subsets of the two-element set $\mathbb{Z}/2\mathbb{Z}$. Each must therefore be a singleton, and they must be the two different residue classes. $\square$

The lemma deliberately excludes vacuous cases. If there are no gray nodes, ordinary and modulo-2 separation are vacuous, whereas the uniform residue form still requires all unpaired levels to have one parity; the analogous issue occurs when there are no unpaired nodes.

3 Prior results and main theorem

The following prior results frame the theorem.

Theorem 3.1 (Haleš et al.; separated coloring implies designability [8]). *A target whose tree representation admits a proper separated coloring is designable in the four-letter Watson–Crick base-pair model.*

Haleš et al. also showed that saturated targets are designable over the four-letter alphabet precisely when their maximum paired degree is at most four, and that the motifs m_5 and $m_{3\bullet}$ are local obstructions. Their separated-coloring theorem is sufficient but not necessary in general.

Theorem 3.2 (Boury–Bulteau–Ponty). *Every $(m_5, m_{3\bullet})$ -free target whose helices all have length at least three admits a modulo-2 separated coloring and can be designed in linear time [3, 4]. Deciding separability remains NP-complete for motif-free targets without isolated base pairs [4].*

The same work gives a motif-free nonseparable construction with minimum helix length 2 that uses many isolated stacks. A documented primary-source search through 19 August 2026 located no theorem or counterexample for the class with at most two isolated stacks; this negative search result is not by itself evidence of novelty.

Theorem 3.3 (Main coloring theorem). *Every target $T \in \mathcal{K}_{\leq 2}$ admits a proper modulo-2 separated coloring. In fact, the coloring has the uniform residue form: all unpaired nodes have one parity and all gray paired nodes have the other.*

Theorem 3.4 (Main design theorem). *Every target $T \in \mathcal{K}_{\leq 2}$ is designable in the four-letter Watson–Crick base-pair model with $\theta = 0$.*

Theorem 3.3 is the new structural result. Theorem 3.4 follows from the sequence construction and uniqueness argument in Sections 9–12.

Proposition 3.5 (Three isolated stacks can prevent modulo-2 separation). *The motif-free target*

$$((.))(.(.))(.)$$

has three isolated stacks and admits no proper modulo-2 separated coloring. It nevertheless admits a proper ordinarily separated coloring and is uniquely designable.

Proof. Suppose a proper modulo-2 separated coloring existed. The target has three unpaired nodes. Its degree-3 root cannot have only black and white children, because properness permits at most one of each, so at least one root child is gray. Lemma 2.9 therefore supplies residue classes ξ for all unpaired nodes and η for all gray pairs.

Each root-child helix has length 2 and terminates in a loop containing an unpaired node. Lemma 4.2 forces that terminal pair to lie at ξ and to be non-gray. The complete length-2 table in Lemma 6.1 is now a necessity statement: at a ξ entry, only the words BB and WW can exit at ξ , so every root child begins non-gray; at an η entry, only GB and GW can exit at ξ , so every root child begins gray. In the first case the root would need three non-gray first colors, although properness permits only one black and one white. In the second it would need three gray first colors, although properness permits at most two. Both cases contradict properness.

For the caveat, the coloring with helix words BB, WW, and GB has gray level set $\{0\}$ and unpaired level set $\{-2, 1, 2\}$, so it is proper and ordinarily separated but not modulo-2 separated. The color-to-sequence construction gives GGACCCAGGAGACU. The exact optimum-and-count dynamic program recorded with the release artifact finds target pair count 6, optimum 6, and optimum count 1. $\square$

Thus the bound two is genuine for the modulo-2 coloring criterion used in the theorem, but not for ordinary separation or designability. The proposition does not claim that every target with three isolated stacks fails modulo-2 separation.

4 Loop types and required residues

Call the innermost pair of a maximal helix its terminal pair. The children of this pair describe the loop at the end of the helix.

Lemma 4.1 (Loop types at a helix terminus). *The loop at the terminal pair of every nonroot helix belongs to exactly one of the following types:*

- *E: an empty terminal pair with no child (possible because $\theta = 0$);*
- *L: a hairpin, bulge, or internal-loop type with at least one unpaired position and zero or one outgoing helix;*
- *M: a multiloop with no unpaired position and two or three outgoing helices.*

The parent of the outer pair of a maximal helix is either the root, an L node with one outgoing helix, or an M node.

Proof. A terminal pair with one paired child and no unpaired child would continue the same helix by the helix-chain lemma, contradicting maximality. If it has an unpaired child, the $m_{3\bullet}$ bound gives at most one paired child. If it has no unpaired child, the m_5 bound permits zero, two, or three paired children. The same maximality argument applies to the parent of a helix head. $\square$

The loops of types *L* and *M* impose parity constraints in any coloring in the uniform residue form.

Lemma 4.2 (Residues at terminal loops). *Fix residues ξ and $\eta = 1 - \xi$ in a proper coloring in the uniform residue form.*

1. *An L loop lies at residue ξ ; its own color and the colors of its paired children are non-gray.*
2. *An M loop lies at residue η .*
3. *An E loop has no forced residue.*

Proof. An unpaired child of an L loop has the same level as the loop, so the loop lies at ξ . A gray loop node or gray child would then place a gray paired node at ξ , contradicting the uniform residue condition. For an M loop, the exposed multiset has size at least three. Properness permits at most one black and one white incidence, so some incidence is gray. If the closing incidence is gray, the loop color is gray and its level is η ; if a child incidence is gray, the child has the same inclusive residue as the loop, again forcing η . $\square$

Thus an L terminal requests exit residue ξ , an M terminal requests η , and an E terminal may be assigned either residue. We choose ξ for E .

5 Feasible entry states for helix subtrees

5.1 Counting isolated stacks in a helix subtree

Contract each maximal helix to one node in the tree of helix subtrees. For a maximal helix H , let $s(H)$ be the number of isolated stacks (maximal helices of length 2) in the complete paired subtree rooted at H . Let K range over the outgoing child helices at the terminal loop of H .

Lemma 5.1 (Subtree decomposition of the isolated-stack count). *For every maximal helix H ,*

$$s(H) = \mathbf{1}_{\{|H|=2\}} + \sum_{K \text{ outgoing from } H} s(K).$$

The child subtrees contain disjoint sets of maximal helices.

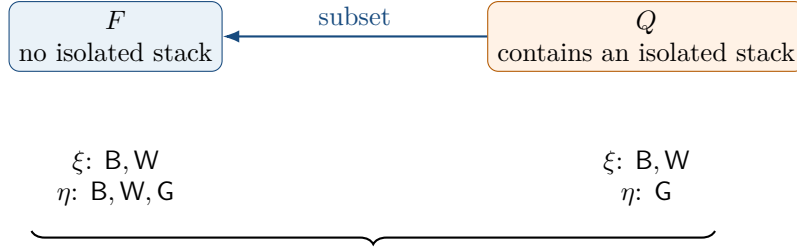
Proof. Every maximal helix in the paired subtree of H is either H itself or lies below exactly one paired child of the terminal pair. The maximal-helix partition and the disjointness of sibling intervals make the sum disjoint. $\square$

Corollary 5.2 (Consequences of the global bound). *If the whole target lies in $\mathcal{K}_{\leq 2}$, then:*

1. *$s(H) \leq 2$ for every helix subtree;*
2. *at most two outgoing child subtrees of any loop contain an isolated stack;*
3. *if two outgoing child subtrees contain isolated stacks, then H is not an isolated stack and hence $|H| \geq 3$.*

The same bound holds among the root-child helix subtrees.

Proof. The first two claims follow from the global count bound and disjointness. If two children already contain at least two isolated stacks and H also had length 2, Lemma 5.1 would give $s(H) \geq 3$. The target-class definition then gives $|H| \geq 3$. $\square$



A child covered only by the Q guarantee contains at least one isolated stack. Since the whole target contains at most two, a loop or root has at most two such children.

Figure 2: The two sets of guaranteed entry states. They differ only at an η entry: Q requires a gray first pair, whereas F also allows black or white.

5.2 Feasible entry states F and Q

Fix ξ and let $\eta = 1 - \xi$. An *entry state* for a helix subtree is a pair consisting of the residue at which the helix is entered and the color of its first pair. Define

$$F = \{\xi B, \xi W, \eta B, \eta W, \eta G\}, \quad Q = \{\xi B, \xi W, \eta G\}.$$

The state ξG is impossible: a gray first pair does not change the entry residue and would therefore lie at ξ , whereas gray nodes must lie at η .

The proof uses the two sets as follows:

- a subtree containing no isolated stack is guaranteed to support every state in F ;
- a subtree containing one or two isolated stacks is guaranteed to support every state in Q .

At a ξ entry, both sets allow black or white. At an η entry, the smaller guarantee Q requires a gray first pair, whereas F also allows black or white.

Definition 5.3 (Feasible entry state). *For a helix subtree H , write $(r, c) \in \text{Accept}(H)$ if the entire subtree can be colored with entry residue r and first-pair color c so that every exposure is proper, every gray pair has residue η , and every unpaired node has residue ξ .*

The proof establishes guaranteed subsets of $\text{Accept}(H)$; it does not claim that F and Q list every feasible coloring.

Theorem 5.4 (Inductive guarantee for helix subtrees). *For every helix subtree H in a target from $\mathcal{K}_{\leq 2}$,*

$$s(H) = 0 \implies F \subseteq \text{Accept}(H),$$

and

$$1 \leq s(H) \leq 2 \implies Q \subseteq \text{Accept}(H).$$

The theorem is proved by well-founded induction in Sections 6–8. Figure 2 summarizes the two guarantees.

6 Local helix transfers

A color word $c_1 \cdots c_h$ on a helix is valid for entry residue ε and requested exit τ if:

1. c_1 is the prescribed first color;
2. every adjacent exposure $\{\text{inv}(c_i), c_{i+1}\}$ is proper;
3. every gray pair occurs at residue η ; and
4. the inclusive residue of the final pair is τ .

When $\tau = \xi$, the construction additionally closes non-gray, as required by L terminals.

6.1 Color patterns for a length-2 helix

Lemma 6.1 (Length-2 transition table). *For a length-2 helix, the complete set of valid color words is:*

<i>entry</i> $\rightarrow$ <i>exit</i>	<i>valid words</i>
$\xi \rightarrow \xi$	BB, WW
$\xi \rightarrow \eta$	BG, WG
$\eta \rightarrow \xi$	GB, GW
$\eta \rightarrow \eta$	BB, WW, GG

Proof. The residues satisfy

$$\ell(x_1) \equiv \varepsilon + \delta(c_1), \quad \ell(x_2) \equiv \ell(x_1) + \delta(c_2) \pmod{2}.$$

Require every gray occurrence to lie at η , the final residue to be τ , and $\{\text{inv}(c_1), c_2\}$ to satisfy the properness capacities. Checking the nine color pairs gives exactly the table. $\square$

In particular, an $\eta \rightarrow \xi$ isolated stack must start gray, and a $\xi \rightarrow \eta$ isolated stack must end gray.

6.2 Long-helix transfer

Lemma 6.2 (Long transfer). *Let $h \geq 3$. Suppose the first color is non-gray, or is gray at an η entry. For either requested exit residue $\tau \in \{\xi, \eta\}$, there is a valid color word of length h preserving the first color; if $\tau = \xi$, the final color can be chosen non-gray.*

Proof. Fix a non-gray color $a \in \{B, W\}$. The local patterns

$$(a, a), \quad (a, G), \quad (G, a), \quad (G, G)$$

are proper, whereas $(a, \text{inv}(a))$ is not. If the first color is a , compare the constant word a^h with a word having one gray insertion at the second or third position, chosen so that the gray lies at η . The two words have opposite exit parities because a non-gray contributes 1 modulo two and a gray contributes 0. The insertion position exists because $h \geq 3$, and the ξ -exit choice cannot close gray. If the first color is gray at an η entry, use G^h for exit η and $G^{h-1}a$ for exit ξ . $\square$

This sufficient transfer is the local engine in the all-long theorem of Boury et al. [3, 4]. The at-most-two proof needs one additional option that was not required by the one-short construction.

6.3 An additional non-gray-ending transfer

Lemma 6.3 (Strengthened long transfer). *Let $h \geq 3$ and suppose the entry state lies in Q . There exists a valid length- h word that exits at η and closes non-gray.*

Proof. At a ξ entry, let $a \in \{B, W\}$ be the prescribed non-gray first color. At an η entry, membership in Q prescribes the first color G; here a denotes the free trailing non-gray color, which we fix as B to make the construction deterministic. Use:

entry and first color	parity of h	word
ξ , first a	odd	a^h
ξ , first a	even	aGa^{h-2}
η , first G	odd	Ga^{h-1}
η , first G	even	GGa^{h-2}

Each word preserves the first color and closes with a . Counting non-gray contributions modulo two gives exit η . The displayed gray positions remain at η , and every adjacency is one of the proper patterns above. In particular, the odd η -entry words at length three are GBB and GWW. $\square$

The words GBB and GWW are the additional local patterns needed for two isolated stacks. They allow a helix to enter at η with a gray first pair, reach a multiloop at the same residue, and still end with a non-gray pair.

7 Color assignments at loops and the root

7.1 Assignments at terminal loops

Let r be the number of outgoing child helix subtrees that contain at least one isolated stack. By Corollary 5.2, $r \leq 2$.

For a terminal pair of type E , there is no outgoing child and $X(v) = \{\text{inv}(a)\}$ is proper. For a terminal loop of type L , the current helix exits at ξ and ends in a non-gray color a . If there is one outgoing helix, assign its first pair the same color a , giving exposure $\{\text{inv}(a), a\}$. The child enters at ξ , where both F and Q allow a .

At a multiloop of type M , the number of outgoing helices is $d \in \{2, 3\}$ and the required exit residue is η .

No child subtree contains an isolated stack ($r = 0$). Use the following first-pair colors:

closing color	$d = 2$ child colors	$d = 3$ child colors
B	B, G	B, G, G
W	W, G	W, G, G
G	B, W	B, W, G

Every child subtree is covered by F and therefore supports each displayed state at an η entry.

One child subtree contains an isolated stack ($r = 1$). Assign gray to that child and use the following colors for the other children:

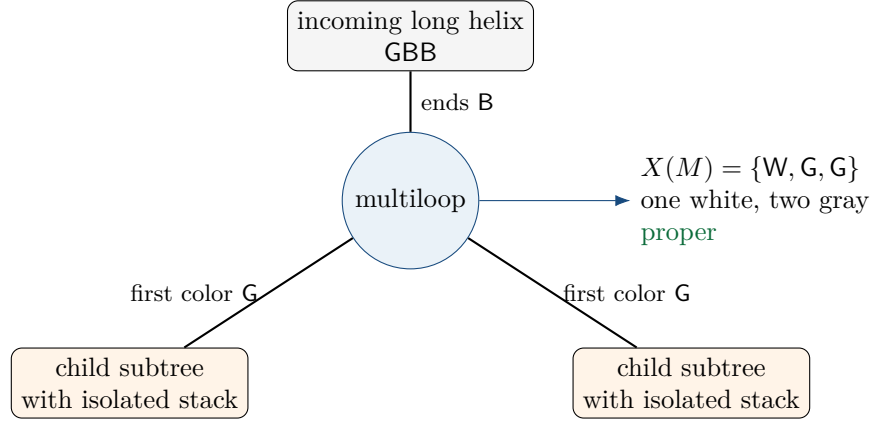


Figure 3: The case with two child subtrees containing isolated stacks. The long incoming helix uses GBB, so it ends black rather than gray. The two child helices can then both start gray without violating properness.

closing color	other color for $d = 2$	other colors for $d = 3$
B	B	B, G
W	W	W, G
G	B	B, W

The child containing an isolated stack supports ηG through Q ; every other child is covered by F .

Two child subtrees contain isolated stacks ($r = 2$). The incoming helix is long by Corollary 5.2. Apply Lemma 6.3 so that it reaches residue η but ends in a non-gray color a . Assign gray to both child helices. If $d = 3$, assign the remaining child, which contains no isolated stack, the color a . The exposures are permutations of

$$\{\text{inv}(a), G, G\} \quad \text{or} \quad \{\text{inv}(a), G, G, a\}.$$

For $a = B$ these are $\{W, G, G\}$ and $\{W, G, G, B\}$; the white case is symmetric. Each exposure has at most one black, one white, and two gray incidences.

Figure 3 depicts the load-bearing $r = 2$ allocation. The table rows are assigned to the actual child helices in backbone order. When $r = 1$ or $r = 2$, the isolated-stack count decomposition identifies exactly which children must receive gray; the remaining colors are assigned deterministically to the other children.

7.2 Assignments at the root

If the root has no paired child, the target is all-unpaired. Choose $\xi = 0$ and use the empty coloring. The root exposure is empty, every unpaired node has level zero, and there are no gray pairs.

Assume the root has at least one paired child. If it has an unpaired child, $m_{3\bullet}$ -freeness gives paired degree at most two and its level forces $\xi = 0$. Assign the child first-pair-color row B at degree one or B, W at degree two. If the root has no unpaired child and paired degree at most two, either naming of the two residues is mathematically valid; to fix all free choices deterministically, choose $\xi = 0$, $\eta = 1$, and use the same child first-pair-color rows.

If the root has no unpaired child and degree $d \in \{3, 4\}$, set $\eta = 0$ and $\xi = 1$. Let r be the number of root-child subtrees that contain an isolated stack; $r \leq 2$. Assign gray to each such child and fill the remaining positions as follows:

root degree	$r = 0$	$r = 1$	$r = 2$
3	B, W, G	G, B, W	G, G, B
4	B, W, G, G	G, B, W, G	G, G, B, W

Each child subtree containing an isolated stack supports ηG through Q , and each remaining child is covered by F .

Lemma 7.1 (Complete local color assignment). *At every terminal loop and at the root, the assignments above give each actual outgoing helix a feasible entry state and keep the actual exposed multiset proper.*

Proof. Lemma 4.1 covers all terminal-loop types. The E and L cases are direct. At a multiloop, r is 0, 1, or 2. The tables cover every possible closing color in the first two cases; Lemma 6.3 supplies a non-gray closing color in the third. The color counts are explicit in every row. The root cases follow from the motif degree bounds and the bound $r \leq 2$. $\square$

8 Global coloring by induction

The local colorings must be combined without allowing choices in one child subtree to affect another. We use the following induction invariant, stated so that it remains valid after the rest of the target is colored.

Definition 8.1 (Subtree coloring invariant). *For a maximal helix H , an entry residue ε , and a first color c_1 , the induction hypothesis gives a coloring defined exactly on the paired nodes in the helix subtree of H . For every total coloring that agrees with it on that subtree and has the specified entry residue, the following statements hold:*

1. *the first pair of H has color c_1 , and the selected color word is used on the actual pairs of H ;*
2. *every paired exposure in the subtree is proper;*
3. *every gray paired node in the subtree has residue η ;*
4. *every unpaired position in the subtree has residue ξ ;*
5. *every outgoing child helix starts with the color assigned at its parent loop; and*
6. *the terminal residue and any required non-gray final color are satisfied.*

The domains of different child subtrees are disjoint, and the pairs in the current helix are disjoint from all child subtrees. Their union is the complete parent subtree. A path from the root to a node in one child does not enter a sibling subtree, so colors in a sibling do not change that node's level. The child subtrees interact only through the exposed multiset at their common parent, which is fixed before the recursive calls.

Proof of Theorem 5.4. Proceed by well-founded induction on the number of paired nodes in the helix subtree of H . Fix ξ , set $\eta = 1 - \xi$, and assume the requested entry state lies in F when $s(H) = 0$, or in Q when $s(H) > 0$.

Let v be the terminal pair of H . By Lemma 4.1, the loop at v is of type E , L , or M . Request exit residue ξ for E and L , and η for M .

If $|H| = 2$, use the exact table in Lemma 6.1. Such a helix belongs to a subtree containing an isolated stack, so the entry state is covered by Q . If $|H| \geq 3$, use Lemma 6.2, except when v is a multiloop with two child subtrees containing isolated stacks. In that case use Lemma 6.3 to reach η with a non-gray final pair.

Input: helix H ; residues ξ, η ; entry residue; first color; count $s(H) \leq 2$
Output: a coloring satisfying the subtree induction invariant
 classify the terminal loop as E , L , or M
 request exit ξ for E/L , and η for M
 identify the child subtrees that contain isolated stacks
if H is long, the terminal is M , and two child subtrees contain isolated stacks **then**
 | choose the additional η -exit transfer with a non-gray final pair
else
 | choose the ordinary length-2 or long-helix transfer
end
 assign first-pair colors to the actual child helices
foreach outgoing child helix K **do**
 | select F or Q according to $s(K)$
 | recursively color K using its assigned first color
end
 combine the color word on H with the disjoint child colorings
 return the coloring and the induction invariant

Algorithm 1: Coloring a helix subtree

Apply Lemma 7.1 to assign the first-pair colors of the child helices. Each child subtree contains fewer paired nodes than the parent. Its isolated-stack count determines whether the required entry state comes from F or Q , and the local assignment lemma proves that the selected color is allowed. Apply the induction hypothesis to each child.

Finally combine the color word on H with the colorings of the disjoint child subtrees. Internal helix exposures are proper by the selected word, the terminal exposure is proper by the local assignment, and all child exposures are proper by induction. The helix transfer and child hypotheses put every gray pair at η . Unpaired positions occur only at loops of type L and therefore have residue ξ . The combined coloring satisfies the stated invariant under every extension outside the subtree. This proves the F guarantee when $s(H) = 0$ and the Q guarantee when $1 \leq s(H) \leq 2$. $\square$

Proof of Theorem 3.3. If the root has degree zero, use the empty coloring. Otherwise choose one of the root assignments above. For each root-child helix, determine whether its subtree contains an isolated stack, take the corresponding state from F or Q , and apply Theorem 5.4. The root-child paired subtrees are disjoint and cover all target pairs, so their colorings combine into a total coloring.

The root exposure is proper by construction. Every nonroot exposure is proper by the induction invariant of its unique subtree. Every gray pair has residue η . Every nonroot unpaired position lies below a loop of type L and has residue ξ ; a root-unpaired position has level zero, and in that case the root construction chooses $\xi = 0$. Therefore the coloring is proper and has the uniform residue form of modulo-2 separation. $\square$

8.1 Algorithm and complexity

The proof gives a deterministic recursive algorithm after the tree representation, maximal helices, child order, isolated-stack counts, root residue convention, symmetric non-gray choices, and gray-pair orientations have all been fixed as above. One version is shown in Algorithm 1.

With adjacency lists and helix membership stored explicitly, each paired node, unpaired node, helix, and child edge is processed $O(1)$ times. Isolated-stack counts are computed bottom-up and colors are assigned top-down. Thus the construction can be implemented in $O(n)$ time and $O(n)$ space. The

Lean development proves existence and correctness and uses classical choice for several finite selections; it is not presented as extracted executable code.

9 From a coloring to an RNA sequence

Let T have a proper coloring. Construct a sequence w top-down:

1. every target-unpaired position receives A;
2. a black pair $[i, j]$ receives $(w_i, w_j) = (G, C)$;
3. a white pair receives $(w_i, w_j) = (C, G)$;
4. a gray pair receives either (A, U) or (U, A) , with the following orientation rule:
 - at the root or below a non-gray parent, gray siblings receive distinct left letters from $\{A, U\}$ in backbone order;
 - below a gray parent, its at most one gray child copies the parent's left letter;
 and every right endpoint is the complement of its left endpoint.

Properness makes the rule well-defined: the root or a non-gray parent has at most two gray children, while a gray parent has at most one gray child.

Lemma 9.1 (Local nucleotide distinctness). *At the root, the left letters of paired children are pairwise distinct. At every nonroot paired node, the left letters of paired children are pairwise distinct and differ from the parent's right letter.*

Proof. For a black parent, properness permits at most one black child and no white child; possible child-left letters are therefore G, A, U, while the parent right letter is C. The white case is symmetric. For a gray parent with left letter $L \in \{A, U\}$, properness permits at most one black, one white, and one gray child. Their possible left letters are G, C, L, while the parent right letter is $\text{comp}(L)$. At the root, the possible child-left letters are G, C, A, U, with distinct orientations assigned to gray siblings. $\square$

The copying rule below a gray parent is essential. For the nested all-gray target $(())$, copying yields AAUU, whereas the opposite orientation gives AUAU, which is also compatible with the disjoint structure $()()$ with the same pair count.

10 Nucleotide-count bound

Let g be the number of gray target pairs, q the number of non-gray target pairs, and u the number of target-unpaired positions.

Lemma 10.1 (Nucleotide-count bound). *The assigned sequence satisfies*

$$\#U = g, \quad \#G = \#C = q, \quad \#A = g + u.$$

Every compatible noncrossing structure S on the same sequence satisfies

$$|S| \leq g + q = |T|.$$

If equality holds, every U, C, and G position is paired in S .

Proof. Each gray target pair contributes one A and one U; each black or white target pair contributes one G and one C; each target-unpaired position contributes one A. Every A–U pair in S consumes a distinct U, and every C–G pair consumes a distinct C. Hence

$$|S| \leq \#U + \#C = g + q = |T|.$$

If equality holds, all U and all C positions are used. Every used C has a distinct G partner, and $\#G = \#C$, so every G is used as well. $\square$

Lemma 10.1 proves that the target has the maximum possible number of base pairs. It does not yet exclude a distinct fold with the same number of pairs.

11 Uniqueness on the paired restriction

This section proves a general uniqueness theorem for the paired restriction of a target, using the local-distinctness condition of Lemma 9.1. It is independent of the short-helix count.

11.1 Saturable sequences and atomic saturable designs

A finite nucleotide word is *saturable* if it admits a compatible noncrossing perfect matching. The empty word is saturable. A nonempty saturable sequence is *atomic* if no nonempty proper prefix is saturable. Following Haleš et al., an *atomic saturable design* is an atomic saturable sequence together with a saturated target for which it is the unique compatible noncrossing perfect matching.

Lemma 11.1 (Adjacent cancellation and suffix cancellation). *A word is saturable if and only if it can be reduced to the empty word by repeatedly deleting adjacent complementary letters. Consequently, if xy and x are saturable, then y is saturable.*

Proof. Suppose a word has a compatible noncrossing perfect matching. Choose a matched pair of minimum span. If its endpoints were not adjacent, the position immediately inside the left endpoint would have a mate strictly inside the chosen pair, producing a smaller span. Therefore the chosen endpoints are adjacent and complementary. Delete them, compress the remaining positions, and induct.

Conversely, reverse a sequence of adjacent complementary deletions. Reinsert each deleted adjacent pair and match its two positions. The new pair is disjoint from or nested within every previously restored pair, so the resulting perfect matching is compatible and noncrossing.

For the suffix statement, define the reduced stack normal form $R(z)$ by scanning left to right and cancelling the next letter whenever it complements the stack top. Adjacent complementary deletion preserves R , and a word reduces to the empty word exactly when $R(z)$ is empty. The stack identity

$$R(xy) = R(R(x)y)$$

then gives $R(y) = R(xy) = \emptyset$ whenever $R(x) = \emptyset$. Thus y is saturable. $\square$

Equivalently, one may encode A, U as inverse generators and C, G as another inverse pair in a free group; suffix cancellation is then ordinary group cancellation. The stack formulation above is sufficient for the proof.

Lemma 11.2 (Atomic endpoint characterization). *A nonempty saturable word z is atomic if and only if every compatible noncrossing perfect matching of z contains the outer pair joining the first and last positions.*

Proof. If z is atomic and the first position is paired to a position before the end, noncrossingness and saturation make the enclosed prefix a nonempty proper saturable prefix. Conversely, if a nonempty proper prefix p is saturable, write $z = ps$. Lemma 11.1 makes s saturable, and concatenating perfect matchings of p and s yields a perfect matching of z omitting the outer pair. $\square$

11.2 Concatenation and wrapping

Lemma 11.3 (Concatenating atomic saturable designs). *Let $z_1, \dots, z_t$ be atomic saturable designs, with $t \geq 1$, and suppose their first letters are pairwise distinct. Then the concatenation $z_1 \cdots z_t$ has the concatenation of the target matchings as its unique compatible noncrossing perfect matching.*

Proof. Let P be a compatible noncrossing perfect matching of the concatenation. If P contains the outer pair of every block, those pairs isolate the blocks and uniqueness holds blockwise.

Otherwise choose the leftmost block z_i whose first position is not paired to its own last position. Earlier blocks are isolated. Let the first position of z_i pair to a position in block z_j , $j \geq i$. If it pairs to the last position of z_j , then $j > i$ and unique complementation forces the first letters of z_i and z_j to be equal, contradicting pairwise distinctness. If it pairs inside z_j , the enclosed word

$$z_i z_{i+1} \cdots z_{j-1} z_j[1, k]$$

is saturable. Repeated suffix cancellation removes the complete saturable blocks $z_i, \dots, z_{j-1}$, leaving a nonempty proper saturable prefix of the atomic word z_j , again a contradiction. $\square$

Lemma 11.4 (Wrapping atomic saturable designs). *Let $z_1, \dots, z_t$ be atomic saturable designs, possibly with $t = 0$. Let a, b be complementary letters. Suppose the first letters of the z_i are pairwise distinct and none equals b . Then*

$$W = a z_1 \cdots z_t b$$

is an atomic saturable design for the target consisting of the outer pair (a, b) around the concatenated child targets.

Proof. For $t = 0$, the two-letter complementary word is immediate. Assume $t \geq 1$. The displayed target proves saturability. Suppose W has a nonempty proper saturable prefix and choose one of minimum length. Its length cannot be one plus the total length of complete initial child blocks, because those lengths are odd while a perfectly matched word has even length. Hence the prefix ends strictly inside some z_j .

In a perfect matching of this shortest prefix, the initial letter a must pair to the final position; otherwise the prefix ending at its partner would be shorter and saturable. The final letter is therefore b . Removing this outer pair leaves

$$z_1 \cdots z_{j-1} z_j[1, k-1]$$

saturable. Repeated suffix cancellation removes the complete blocks, so $z_j[1, k-1]$ is saturable. Atomicity forces $k-1 = 0$, making the first letter of z_j equal to b , a contradiction. Therefore W is atomic. Lemma 11.2 forces the outer pair in every perfect matching, and Lemma 11.3 gives uniqueness inside it. $\square$

Theorem 11.5 (Saturated local-distinctness theorem). *Let R be a saturated target and z a compatible sequence. Suppose that:*

1. *at the root, the left letters of paired children are pairwise distinct; and*
2. *at every nonroot pair $[i, j]$, the left letters of paired children are pairwise distinct and none equals z_j .*

Then R is the unique compatible noncrossing perfect matching of z .

Proof. Induct upward through the interval tree. A leaf pair is Lemma 11.4 with $t = 0$. At a nonroot pair $[i, j]$, saturation makes the interior a concatenation of paired-child subwords. By induction these are atomic saturable designs. Compatibility makes z_i and z_j complementary, and the local hypotheses are precisely those of Lemma 11.4; hence the whole interval subword is an atomic saturable design. At the root, the complete word is a concatenation of root-child atomic saturable designs with pairwise distinct first letters, so Lemma 11.3 gives uniqueness. $\square$

Lemma 9.1 shows that the sequence produced from every proper coloring satisfies the hypotheses of Theorem 11.5 on its paired restriction.

12 Excluding alternative optimal folds

It remains to show that a different compatible structure with the same number of pairs cannot use a position that is unpaired in the target.

12.1 G–C prefix balance and levels

For the constructed sequence w , define the inclusive prefix balance

$$\Lambda(k) = \#G \text{ in } w[1, k] - \#C \text{ in } w[1, k], \quad \Lambda(0) = 0.$$

Lemma 12.1 (Levels equal prefix balances). *For every target pair $v = [i, j]$,*

$$\Lambda(i) = \ell(v).$$

For every target-unpaired position k , $\Lambda(k)$ equals the level of its unpaired interval-tree node. If $v = [i, j]$ is gray, then

$$\Lambda(j) = \Lambda(i) = \ell(v).$$

Proof. Every complete target subtree contains equal numbers of G and C: a black or white pair contributes one of each, while gray pairs and unpaired A positions contribute neither. Completed earlier siblings therefore contribute zero to the running balance. At a paired left endpoint, the only uncanceled nonzero contributions are the left endpoints on the open ancestor path. Black contributes +1, white -1 , and gray 0, exactly reproducing the inclusive level. An unpaired A changes nothing and inherits the enclosing level. A gray pair and its complete interior contribute zero, so its right and left endpoints have the same balance. $\square$

Lemma 12.2 (Level-imbalance obstruction). *Let S be a compatible noncrossing structure of w . If S contains an A–U pair between positions $a < b$ with $\Lambda(a) \neq \Lambda(b)$, then S leaves at least one G or C position unpaired.*

Proof. No position strictly between a and b can pair outside that interval without crossing (a, b) . Since the endpoints have zero G-minus-C weight,

$$\#G \text{ in } (a, b) - \#C \text{ in } (a, b) = \Lambda(b) - \Lambda(a) \neq 0.$$

If every interior G and C were paired, compatibility and noncrossing closure inside the arc would pair them bijectively, forcing equal counts. Thus some G or C remains unpaired. $\square$

12.2 Uniqueness among optimal folds

Theorem 12.3 (No distinct optimal structure). *Let T have a proper separated coloring and let w be the sequence assigned in Section 9. If a compatible noncrossing structure S satisfies $|S| = |T|$, then $S = T$.*

Proof. By Lemma 10.1, equality in pair count requires every U, C, and G position to be paired.

Suppose a position i unpaired in T is paired to j in S . Then $w_i = A$ and compatibility gives $w_j = U$. Every U belongs to a gray target pair v . Lemma 12.1 identifies $\Lambda(i)$ with the target-unpaired level and $\Lambda(j)$ with the gray-pair level, regardless of which endpoint of v carries U. Separation makes these levels different. After ordering the endpoints as $a = \min\{i, j\}$ and $b = \max\{i, j\}$, Lemma 12.2 says that S leaves a G or C unpaired, contradicting the equality case of Lemma 10.1. Hence every target-unpaired position remains unpaired in S .

The two structures have equal pair count and therefore equal numbers of unpaired positions. Their unpaired-position sets are equal. Delete that common set and compress the remaining positions in backbone order. The restrictions of T and S are saturated, compatible, and noncrossing on the same restricted word. Deleting target-unpaired singleton nodes does not change paired parent-child relations, so Lemma 9.1 transfers to the restricted target. Theorem 11.5 makes that target the unique compatible perfect matching. Thus the restrictions agree, and restoring the common unpaired positions gives $S = T$. $\square$

Proof of Theorem 3.4. Theorem 3.3 gives a proper coloring in the uniform residue form, hence a proper separated coloring. Construct the sequence in Section 9. Lemma 10.1 shows that no compatible structure has more pairs than the target, and Theorem 12.3 shows that no distinct compatible structure can attain the same pair count. Therefore the target is the unique compatible noncrossing structure with the maximum number of base pairs and hence the unique minimum-energy fold. $\square$

13 Worked examples

The universal proof does not depend on finite examples, but concrete instances clarify the two new coloring cases.

13.1 Two isolated stacks at a degree-three root

Consider

$$T_1 = (())((()))((())),$$

which has two isolated stacks at the root and one root helix of length three. The root has no unpaired position and degree three, so choose $\eta = 0$, $\xi = 1$, and the root row for two such children

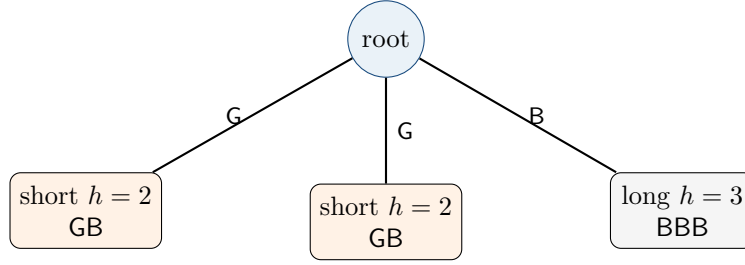
$$G, G, B.$$

Each isolated stack enters at η through gray and terminates at E , so it uses the $\eta \rightarrow \xi$ word GB. The long helix enters through black and uses a length-three transfer to its E terminal. One deterministic orientation gives

$$w_1 = AGCUUGCAGGGCCC.$$

The target has seven pairs. Exact Nussinov dynamic programming [11] gives optimum score seven and optimum count one.

Figure 4 makes the degree-3 root assignment explicit: the two gray first-pair colors are assigned to the two isolated-stack subtrees.



Target: $((()))((()))((()))$ Sequence: AGCUUGCAGGGCCC
 root exposure $\{G, G, B\}$ is proper; optimum pair count 7, unique optimum.

Figure 4: A degree-three root with two child subtrees containing isolated stacks. The root row assigns gray to the two child subtrees containing isolated stacks and black to the remaining long-helix subtree.

13.2 Two isolated-stack subtrees at an internal multiloop

The target

$$T_2 = (((((((((()))((())))))((())))))$$

has five maximal helices with lengths $(3, 3, 2, 2, 3)$. Its outer length-3 helix ends at an M loop with two long children. One of those long children ends at a second M loop whose two children are the two isolated stacks. At that internal loop, both child subtrees contain an isolated stack, so the incoming length-3 helix uses the strengthened transfer

GBB

and closes black. The two short children each receive gray and use GB. The root has paired degree one and no unpaired child, so the deterministic small-root convention fixes

$$\xi = 0, \quad \eta = 1.$$

The helix words, in the order outer, critical, two short children, and side child, are

BBB, GBB, GB, GB, BBB,

with integer levels $(1, 2, 3)$, $(3, 4, 5)$, $(5, 6)$, $(5, 6)$, and $(4, 5, 6)$, respectively. The deterministic orientation rule then gives

$$w_2 = \text{GGGAGGAGCUUGCACCUGGGCCCCC}.$$

The target has thirteen pairs; exact Nussinov dynamic programming gives optimum score thirteen and optimum count one.

The final review also reported GGGGGAGGCCCCGGUCCAGGCCUCCC. That sequence is compatible and uniquely optimal for the unchanged target, but its helix words are BBB, BBG, BB, WW, and GBB. It therefore does not instantiate the stated F/Q recursion or the internal strengthened-transfer case. We use the independently reconstructed sequence above, which does.

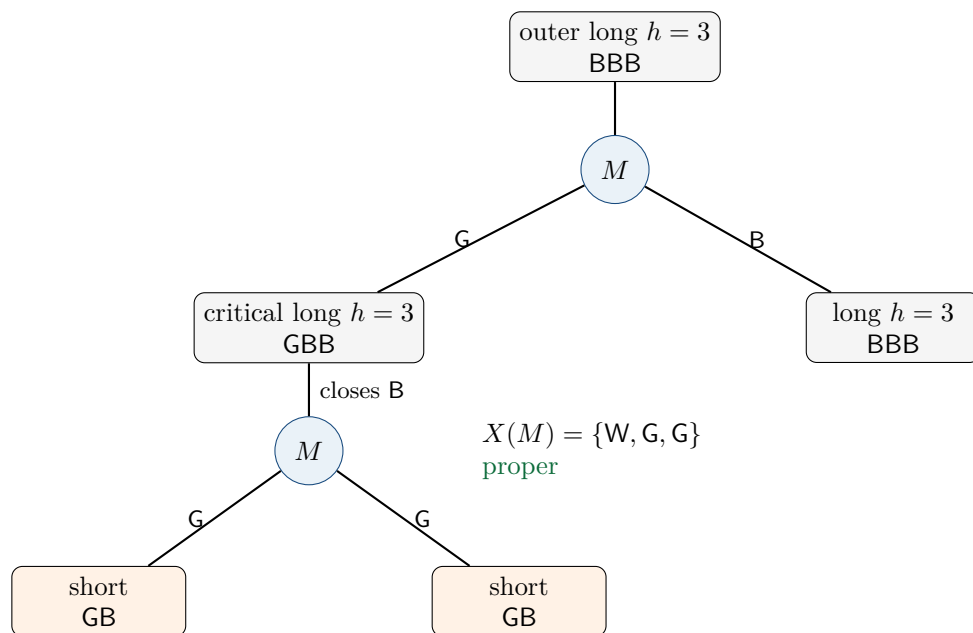


Figure 5: An internal multiloop with two child subtrees containing isolated stacks. The word **GBB** preserves the gray entry, reaches the M residue, and closes non-gray, allowing both child helices to start gray.

Figure 5 shows the exact internal $r = 2$ configuration realized by this deterministic coloring.

13.3 What the examples do and do not show

The examples demonstrate that the new $r = 2$ root and multiloop cases are nonvacuous. Their exact folding checks are regression tests, not evidence for the universal theorem. Universality comes from the induction over arbitrary helix subtrees.

14 Lean formalization and artifact verification

The complete theorem was formalized in Lean 4 [10]. The following listing is generated directly from the frozen source; it includes the exact public proposition, theorem, and proof term rather than Lean-like pseudocode.

Listing 1: Exact final Lean proposition and theorem, extracted from the frozen source.

```
def AtMostTwoShortHelixDesignabilityStatement : Prop :=
  ∀ {n : Nat} (T : SecondaryStructure n),
    InTargetClassKLeTwo T →
      ∃ w : Sequence n, UniqueDesigns w T
theorem atMostTwoShortHelixDesignability :
  AtMostTwoShortHelixDesignabilityStatement := by
  intro n T hK
  exact ⟨sequenceOfProperColoring
    (globalColoringCertificateLeTwo hK).coloring
    (globalColoringCertificateLeTwo hK).proper,
    atMostTwoShortHelices_uniqueDesigns⟩hK
```

The definition of `UniqueDesigns` quantifies over every `SecondaryStructure n` compatible with the same complete sequence and uses a strict pair-count inequality for every distinct competitor. The target class uses the actual cardinality of the finite set of maximal helices of length 2. A checked source map records the source path, Lean module, inclusive line range, full-source digest, and extracted-block digest of every declaration displayed here and in Appendix B. A generated audit module imports the real theorem module and successfully applies `#check` to all 28 displayed declarations.

14.1 Structure of the Lean development

The Lean development follows the same sequence of ideas as the paper:

1. the four-letter alphabet, complementarity, arbitrary noncrossing partial matchings, interval tree, motifs, maximal helices, and the public unique-designability statement;
2. colors, exposures, levels, the uniform modulo-2 condition, exact helix transfers, and the decomposition of the target into disjoint helix subtrees;
3. the definitions of the feasible states F and Q , the isolated-stack count decomposition, the strengthened non-gray-closing transfer, color assignments to the actual child helices, and the recursive subtree invariant;
4. deterministic sequence assignment, followed by the counting and uniqueness lemmas; and
5. composition of the new global coloring with the general coloring-to-design theorem.

The old exact-one theorem is not a premise of the at-most-two result; it is derived afterward as a corollary.

14.2 Concrete semantic regression theorems

Two new downstream-only Lean modules pin the formal definitions to concrete publication examples without entering the dependency closure of Listing 1. Kernel-checked theorems establish that T_1 and the corrected T_2 belong to `InTargetClassKLeTwo`; that their displayed literal sequences satisfy `UniqueDesigns`; that the T_2 coloring has helix words BBB/GBB/GB/GB/BBB and satisfies the named-residue condition with $\xi = 0$; that the three-isolated-stack and length-1 examples are rejected by

the target predicate; and that AUAU fails to uniquely design the nested target $(())$. Ordinary kernel reduction over all $3^6 = 729$ colorings also proves that the three-isolated-stack target has no coloring that is both proper and strongly modulo-2 separated.

The literal design theorems do not call an external folding oracle. They identify each literal word with the generic sequence produced from an explicit proper separated coloring, then apply the universal uniqueness theorem. Their transitive axiom set is the same three foundational principles listed below. Hash comparison confirms that all 60 pre-existing tracked Lean files remained byte-identical and that no reverse import from the theorem closure to either publication-example module was introduced.

14.3 Axiom check and isolated rebuild

Lean reports the following transitive axiom set for the final theorem:

`{propext, Classical.choice, Quot.sound}`.

The final theorem uses no project-defined axiom or unfinished proof placeholder. In particular, its dependencies contain no `sorry`, `admit`, `sorryAx`, mathematical use of `unsafe`, `native_decide`, or external proof program.

A frozen source archive was rebuilt in an isolated Linux container using the pinned Lean and Mathlib revisions. The archived qualification rebuild completed 3,070 jobs, printed the exact final theorem, checked its foundational dependencies, searched the source for unfinished or externally supplied proof steps, and verified that compilation did not alter the source files.

The final publication branch was also qualified by a measured local clean build. The authoritative 46-file theorem closure contains 20,738 physical lines, 16,908 nonblank noncomment code lines, and 1,616 source declaration commands: 390 `def`, 36 `abbrev`, 20 `structure`, 7 `inductive`, 1,102 `theorem`, and 61 `instance` commands. After `lake clean`, the bare full-project command `lake build` passed with 3,070 Lake graph jobs in 894.10 seconds wall-clock on the recorded ten-logical-CPU host. These are measured qualification values, not performance claims. The two downstream publication modules add 613 physical lines and 56 declaration commands; their named build and 18-endpoint axiom audit also passed. The release metrics report states the counting method, environment, user and system times, and complete command log; its SHA-256 is `b97a0465d97c9568211cf959e35569a4f82c766f511ca553fd1dd6ef1e1433f9`.

A separate anonymous package contained only the mathematical specification, the handwritten Lean files needed by the final theorem, and the version-pinned build files. A fresh Claude Code session checked the file list and checksums, built the target module, traced the theorem dependencies, tested concrete examples and counterexamples, and compared the Lean definitions with the specification. This was a blinded automated audit by an independent AI session, not an independent human review. Its recorded verdict was **FAITHFUL AND COMPLETE**. The audit report records the following checks:

- exactly four nucleotides and strict Watson–Crick compatibility;
- arbitrary noncrossing partial-match competitors on the same sequence;
- the exact at-most-two target class, including the all-unpaired case;
- the isolated-stack count decomposition and the inductive guarantees expressed by F and Q ;
- the $h = 3$ and $h = 4$ strengthened transfers;
- loop and root color assignments when two child subtrees contain isolated stacks;
- that subtree colorings remain valid when the rest of the target is colored, and that every recursive call is made on a smaller subtree; and

- the direct dependency of the final theorem on the new inductive coloring construction.

Validated release artifact identifiers.

- canonical source archive: b077b6118a6cbfadb6c6fce00af65ed34125ffa431f6ca8b7e2e9de3c268d40f;
- release-qualification results package: 2da05881fb3808e7add46526f6f202f81e6f0b4f9b37acd622a8acd7b435615b;
- blinded fidelity bundle: bfd2c8d90f5666ccb2dda6a3e9c2f81b84419ae84cc25beb3f880c80b5c66b8b;
- Claude fidelity-audit package: 0a77ea7e9f9dc0342a14330b93644dc943a0b2c0e89f302b092ca08de404092b; and
- manuscript source package: generated from this revision, with its digest recorded in the external release manifest.

Canonical repository: <https://github.com/ajogalekar/rna-at-most-two-short-helices>. Archival DOI: [10.5281/zenodo.22075874](https://doi.org/10.5281/zenodo.22075874). Version 1.0.0 uses Apache License 2.0 for Lean source and software tools, Creative Commons Attribution 4.0 International for manuscript text, original figures, and project documentation, and the STIX Font License / SIL Open Font License 1.1 terms for bundled fonts; `LICENSES.md` in the archive gives the authoritative path scopes. The DOI was reserved before this PDF was built and identifies the immutable Zenodo Software record for version 1.0.0. The external SHA-256 manifest identifies the exact release bytes.

14.4 What formal verification establishes

Kernel acceptance establishes that the final theorem follows from the formal definitions and the listed foundational principles. The isolated rebuild establishes reproducibility of the exact frozen source. The automated specification comparison addresses, with lower evidentiary weight, the separate risk that Lean might have proved a subtly weakened or different theorem. Because that comparison was performed by an AI system, its verdict is not a substitute for kernel checking, reproducibility, public inspection, or human peer review. None of these checks establishes biological realism, novelty, importance, empirical utility, or correctness of claims outside the formal model.

15 Development history and audit trail

The proof was developed with substantial assistance from AI systems, but the mathematical model and acceptance criteria were set by the human author. A recurring difficulty was the distinction among the target structure, the sequence assigned to it, and alternative folds of that same sequence. Recasting the argument in terms of base-pair use and alternative folds helped clarify the role of the coloring construction.

The development proceeded in several stages:

1. clarification of the theorem statement and its quantifiers;
2. generation and revision of candidate proofs by ChatGPT and Codex;
3. independent attempts by Claude Code to find counterexamples or missing cases;
4. independent programs for exact colorability and optimum-counting folding;

5. a staged Lean formalization with source review after each major part; and
6. an isolated rebuild followed by a blinded comparison of the mathematical specification with the Lean source.

Several early versions contained genuine errors or incomplete case analyses, including an incorrect orientation rule for gray pairs and overly broad statements about long-helix transfers. The final theorem is supported first by the mathematical argument and Lean kernel checking, then by the reproducible isolated build. The blinded automated AI audit is reported as a secondary specification cross-check, not as independent human validation. The human author remained responsible for the problem definition, evidence standards, version control, interpretation, and publication claims.

16 Scope, limitations, and discussion

16.1 Model limitations

The theorem is exact only in the stated base-pair model. It excludes G–U wobble, pseudoknots, stacking energies, loop entropies, ionic effects, ensemble objectives, kinetics, and three-dimensional constraints. Minimum arc length is $\theta = 0$, so empty hairpins and adjacent pairs are legal. If a target remains valid under a stricter minimum-loop convention, the same design remains unique because the competitor set only shrinks; however, some targets in $\mathcal{K}_{\leq 2}$ cease to be legal targets when $\theta > 0$.

Avoidance of $m_{3\bullet}$ is biologically restrictive. Any nonroot loop with at least two branch helices must contain no unpaired nucleotide in this model, and an exterior loop with an unpaired nucleotide may have at most two paired children. The theorem therefore concerns a narrow but mathematically natural combinatorial class rather than typical full thermodynamic RNA architectures.

16.2 What the bound two means

The theorem gives a uniform modulo-2 coloring for every target with at most two isolated stacks. Proposition 3.5 shows that a target with three isolated stacks can already fail to admit any proper modulo-2 separated coloring. It does not show that three isolated stacks always imply modulo-2 nonseparability, ordinary nonseparability, or undesignability. Indeed, the displayed target is ordinarily separated and designable. The natural next questions are therefore:

- which targets with three or more isolated stacks admit modulo-2 separated colorings;
- whether a higher modulus, ordinary separation, or biseparability [5] yields broader bounded-isolated-stack theorems;
- whether other finite collections of feasible entry states yield broader guarantees; and
- which structural placements, rather than counts alone, govern the transition from universal guarantees to obstructions.

16.3 Novelty and related-work boundary

The all-long case was previously covered by Boury et al., and the general implication from separated colorings to designability originates with Haleš et al. The new contribution is the direct induction on helix subtrees proving that zero, one, or two length-2 helices can always be accommodated while preserving the stated modulo-2 residue condition. A documented search through 19 August 2026 did not locate an exact prior theorem or counterexample for this bounded-isolated-stack class. That search is necessarily incomplete and should not be treated as a proof of novelty.

16.4 Algorithmic implications

The coloring algorithm runs in linear time when the tree representation and maximal helices are stored explicitly. After fixing the root-residue convention, symmetric transfer choices, child order, and gray-pair orientations, the top-down rules supply a deterministic sequence. This gives an exact constructive baseline for testing inverse-folding software and for initializing more realistic thermodynamic design methods. Whether these sequences are good seeds under Turner energies is an empirical question; published work suggests that separated and biseparable sequences can be useful starting points, but the present theorem makes no thermodynamic claim [4, 5].

17 Conclusion

We have proved that every motif-free RNA target with no isolated base pair, at most two isolated stacks, and all other helices of length at least three is designable in the four-letter Watson–Crick base-pair model. The proof uses the two sets of feasible entry states F and Q in an induction on helix subtrees. If two child subtrees both require gray first pairs, they already contain the two isolated stacks allowed in the target. The incoming helix is therefore long and can reach residue η while ending non-gray, allowing both child helices to start gray.

The complete result, including the sequence construction and uniqueness theorem, has been formalized in Lean and rebuilt from frozen source. A blinded automated AI audit separately compared the mathematical specification with the Lean source as a secondary cross-check. The result gives a new positive class in the $h_{\min} = 2$ regime and an example of how AI-assisted proof development can be brought to a reproducible formal endpoint.

A Local transition and color-assignment tables

A.1 Two-pair transitions

entry $\rightarrow$ exit	valid words
$\xi \rightarrow \xi$	BB, WW
$\xi \rightarrow \eta$	BG, WG
$\eta \rightarrow \xi$	GB, GW
$\eta \rightarrow \eta$	BB, WW, GG

A.2 Multiloop color assignments

child subtrees containing isolated stacks r	closing color	$d = 2$	$d = 3$
0	B	B, G	B, G, G
0	W	W, G	W, G, G
0	G	B, W	B, W, G
1	B	G, B	G, B, G
1	W	G, W	G, W, G
1	G	G, B	G, B, W
2	B	G, G	G, G, B
2	W	G, G	G, G, W

In the $r = 1$ and $r = 2$ rows, gray entries are assigned to the children whose subtrees contain an isolated stack; displayed order is schematic.

A.3 Root color assignments at degree three and four

root degree	$r = 0$	$r = 1$	$r = 2$
3	B, W, G	G, B, W	G, G, B
4	B, W, G, G	G, B, W, G	G, G, B, W

B Selected Lean definitions

The following declarations are exact Unicode excerpts generated from the frozen source, not simplified retypings. Blank lines are the only inserted material between source blocks. The accompanying source map and `#check` audit are distributed with the manuscript source.

Listing 2: Exact core-model declarations extracted from the frozen Lean source.

```

inductive Nucleotide where
| A
| C
| G
| U
deriving DecidableEq, Repr
def comp : Nucleotide → Nucleotide
| A => U
| U => A
| C => G
| G => C
def Compatible (x y : Nucleotide) : Prop := y = x.comp
abbrev Sequence (n : Nat) := Fin n → Nucleotide
structure Arc (n : Nat) where
left : Fin n
right : Fin n
ordered : left < right
deriving DecidableEq
def Crosses (a b : Arc n) : Prop :=
(a.left < b.left ∧ b.left < a.right ∧ a.right < b.right) ∨
(b.left < a.left ∧ a.left < b.right ∧ b.right < a.right)
def IsPartialMatching (P : Finset (Arc n)) : Prop :=
∀ {a : Arc n}, a ∈ P →
  ∀ {b : Arc n}, b ∈ P →
    a.ShareEndpoint b → a = b
def IsNoncrossing (P : Finset (Arc n)) : Prop :=
∀ {a : Arc n}, a ∈ P →
  ∀ {b : Arc n}, b ∈ P →
    ¬ a.Crosses b
structure SecondaryStructure (n : Nat) where
arcs : Finset (Arc n)
isPartialMatching : IsPartialMatching arcs
isNoncrossing : IsNoncrossing arcs
def StructureCompatible (w : Sequence n) (S : SecondaryStructure n) : Prop :=
  ∀ a ∈ S.arcs, Compatible (w a.left) (w a.right)
def pairCount (S : SecondaryStructure n) : Nat := S.arcs.card

```

```

def UniqueDesigns (w : Sequence n) (T : SecondaryStructure n) : Prop :=
  StructureCompatible w T ∧
  ∀ S : SecondaryStructure n,
    StructureCompatible w S →
    S ≠ T →
    pairCount S < pairCount T

```

Listing 3: Exact helix and target-class declarations extracted from the frozen Lean source.

```

abbrev HelixCandidate (n : Nat) := Arc n × Fin (n + 1)
def outer (H : HelixCandidate n) : Arc n := H.1
def length (H : HelixCandidate n) : Nat := H.2.val
def HasStackOffset (T : SecondaryStructure n) (outer : Arc n) (k : Nat) : Prop :=
  ∃ a ∈ T.arcs, outer.StackOffset a k
def IsMaximalHelixRun (T : SecondaryStructure n) (outer : Arc n) (h : Nat) : Prop :=
  0 < h ∧
  outer ∈ T.arcs ∧
  (¬ ∃ a ∈ T.arcs, a.Stacked outer) ∧
  ∀ (k : Fin h, HasStackOffset T outer k.val) ∧
  ¬ HasStackOffset T outer h
def IsMaximalHelix (T : SecondaryStructure n) (H : HelixCandidate n) : Prop :=
  IsMaximalHelixRun T H.outer H.length
abbrev MaximalHelix (T : SecondaryStructure n) :=
  {H : HelixCandidate n // IsMaximalHelix T H}
def outer {T : SecondaryStructure n} (H : MaximalHelix T) : Arc n := H.val.outer
def length {T : SecondaryStructure n} (H : MaximalHelix T) : Nat := H.val.length
def HasM5 : Prop :=
  ∃ p : PairedOrRootNode T, 4 < pairedDegree T p
def HasM3Dot : Prop :=
  ∃ p : PairedOrRootNode T,
    HasUnpairedChild T p ∧ 2 < pairedDegree T p
def lengthTwoHelices (T : SecondaryStructure n) : Finset (MaximalHelix T) :=
  maximalHelicesOfLength T 2
def shortHelixCount (T : SecondaryStructure n) : Nat :=
  (lengthTwoHelices T).card
def InTargetClassKLeTwo (T : SecondaryStructure n) : Prop :=
  shortHelixCount T ≤ 2 ∧
  ∀ (H : MaximalHelix T, H.length ≠ 1) ∧
  ∀ (H : MaximalHelix T, H.length ≠ 2 → 3 ≤ H.length) ∧
  ¬ HasM5 T ∧
  ¬ HasM3Dot T

```

C Formal verification record and artifacts

Artifact	Record
Canonical proof specification	CANONICAL_AT_MOST_TWO_PROOF.md; SHA-256 fb9bee1126791b9ca53b3903c061df27e3856dd20b9c7fc248b7acd4e5326841.
Canonical source archive	SHA-256 b077b6118a6cbfadb6c6f6ce00af65ed34125ffa431f6ca8b7e2e9de3c268d40f.

Artifact	Record
Lean toolchain	Lean 4.34.0-rc1; Mathlib revision de5ce8a9a66a4aa68a9bdbb35b63a06d34d9ca11.
Isolated rebuild	Full clean build passed; final theorem printed with axiom set [propext, Classical.choice, Quot.sound].
Release-qualification results package	SHA-256 2da05881fb3808e7add46526f6f202f81e6f0b4f9b37acd622a8acd7b435615b.
Blinded fidelity bundle	SHA-256 bfd2c8d90f5666ccb2dda6a3e9c2f81b84419ae84cc25beb3f880c80b5c66b8b.
Claude fidelity-audit package	Automated-AI verdict FAITHFUL AND COMPLETE ; SHA-256 0a77ea7e9f9dc0342a14330b93644dc943a0b2c0e89f302b092ca08de404092b.
Publication examples	Downstream modules PublicationExamples.lean and PublicationExamplesAxiomAudit.lean; no reverse import and no change to any pre-existing Lean file.
Publication qualification metrics	Clean full-project build: 3,070 Lake graph jobs and 894.10 seconds wall-clock; closure: 46 files, 20,738 physical lines, 1,616 declaration commands; report SHA-256 b97a0465d97c9568211cf959e35569a4f82c766f511ca553fd1dd6ef1e1433f9.
Derived listings	SOURCE_MAP.tsv, extraction script, and 28-declaration #check audit included in the manuscript source package.
Manuscript source package	Generated from this revision; SHA-256 recorded in the external release manifest to avoid a circular self-digest.
Availability	Version 1.0.0; canonical repository https://github.com/ajogalekar/rna-at-most-two-short-helices ; archival DOI 10.5281/zenodo.22075874; mixed license per LICENSES.md.

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Data and code availability

The version 1.0.0 archival release contains the manuscript source and PDF, source-derived Lean listings and their audit, deterministic example checker and output, citation and provenance audits, downstream Lean publication examples, and frozen qualification packages. Appendix C records the canonical repository, archival DOI, and mixed-license scope; the external SHA-256 manifest identifies the exact release bytes.

References

- [1] M. Andronescu, A. P. Fejes, F. Hutter, H. H. Hoos, and A. Condon. A new algorithm for RNA secondary structure design. *Journal of Molecular Biology*, 336(3):607–624, 2004. doi:10.1016/j.jmb.2003.12.041.
- [2] E. Bonnet, P. Rzażewski, and F. Sikora. Designing RNA secondary structures is hard. *Journal of Computational Biology*, 27(3):302–316, 2020. doi:10.1089/cmb.2019.0420.
- [3] T. Boury, L. Bulteau, and Y. Ponty. RNA inverse folding can be solved in linear time for structures without isolated stacks or base pairs. In S. P. Pissis and W.-K. Sung, editors, *24th International Workshop on Algorithms in Bioinformatics (WABI 2024)*, volume 312 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 19:1–19:23. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 2024. doi:10.4230/LIPIcs.WABI.2024.19.
- [4] T. Boury, S. Gardelle, L. Bulteau, and Y. Ponty. RNA inverse folding can be solved in linear time for structures without isolated stacks or base pairs. *Algorithms for Molecular Biology*, 20:20, 2025. doi:10.1186/s13015-025-00278-6.
- [5] T. Boury, L. Sidl, I. L. Hofacker, Y. Ponty, and H.-T. Yao. Old dog, new tricks: Exact seeding strategy improves RNA design performances. In S. Sankararaman, editor, *Research in Computational Molecular Biology (RECOMB 2025)*, volume 15647 of *Lecture Notes in Computer Science* (sub-series *Lecture Notes in Bioinformatics*), pages 134–152. Springer, Cham, 2025. doi:10.1007/978-3-031-90252-9_9.
- [6] A. Busch and R. Backofen. INFO-RNA—a fast approach to inverse RNA folding. *Bioinformatics*, 22(15):1823–1831, 2006. doi:10.1093/bioinformatics/btl194.
- [7] J. A. García-Martín, P. Clote, and I. Dotú. RNAiFOLD: A constraint programming algorithm for RNA inverse folding and molecular design. *Journal of Bioinformatics and Computational Biology*, 11(2):1350001, 2013. doi:10.1142/S0219720013500017.
- [8] J. Haleš, A. Héliou, J. Mañuch, Y. Ponty, and L. Stacho. Combinatorial RNA design: designability and structure-approximating algorithm in Watson–Crick and Nussinov–Jacobson energy models. *Algorithmica*, 79(3):835–856, 2017. doi:10.1007/s00453-016-0196-x.
- [9] I. L. Hofacker, W. Fontana, P. F. Stadler, L. S. Bonhoeffer, M. Tacker, and P. Schuster. Fast folding and comparison of RNA secondary structures. *Monatshefte für Chemie / Chemical Monthly*, 125(2):167–188, 1994. doi:10.1007/BF00818163.
- [10] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, LNCS 12699, pages 625–635, 2021. doi:10.1007/978-3-030-79876-5_37.
- [11] R. Nussinov and A. B. Jacobson. Fast algorithm for predicting the secondary structure of single-stranded RNA. *Proceedings of the National Academy of Sciences of the United States of America*, 77(11):6309–6313, 1980. doi:10.1073/pnas.77.11.6309.
- [12] D. H. Turner and D. H. Mathews. NNDB: The nearest neighbor parameter database for predicting stability of nucleic acid secondary structure. *Nucleic Acids Research*, 38(suppl. 1):D280–D282, 2010. doi:10.1093/nar/gkp892.
- [13] J. N. Zadeh, B. R. Wolfe, and N. A. Pierce. Nucleic acid sequence design via efficient ensemble defect optimization. *Journal of Computational Chemistry*, 32(3):439–452, 2011. doi:10.1002/jcc.21633.